Tariq Khalaf

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EDUCATION

University of North Carolina at Charlotte *Expected: Dec 2028*
Bachelors of Science in Electrical and Computer Engineering

SKILLS

Design & Simulation: KiCad, Fusion 360, AutoCAD, SolidWorks, PCB Design, Circuit Diagrams, VS Code, Dev-C++, Meshmixer, 3D Printing
Programming: Python, C++, JavaScript, Node.js, Arduino, Embedded C, UART, PWM, I2C/SPI basics
Engineering Skills: Soldering, PCB bring-up/debugging, buck regulators, USB-C power/programming, motor control, circuit prototyping

EXPERIENCE

Volunteer Tutoring – Mathematics *Sep 2023 – Sep 2024*
- Tutored students one-on-one in Algebra and Trigonometry, adapting explanations to individual learning styles.
- Strengthened foundational concepts through practice problems, step-by-step review, and test-prep support.

Volunteer Helper – Introduction to Engineering *Jan 2026 – Jul 2026*
- Assisted students during labs and 15+ projects by explaining engineering concepts and troubleshooting builds.
- Supported hands-on activities involving Arduino, circuits, sensors, mechanical prototypes, and problem-solving workflows.

PROJECTS

Electronic Speed Controller – Academic Project *Dec 2025 – Jan 2026*
- Designed a 24V DC motor ESC supporting up to 5A continuous current, emphasizing safe power delivery and reliability.
- Created multi-layer PCB layouts in KiCad with ground planes, power pours, via stitching, and high-current trace planning.
- Selected parts using datasheets and supplier availability to balance performance, cost, footprint, and manufacturability.

Embedded MCU Power & Programming Board – Personal Project *Jun 2026 – Jul 2026*
- Designed a custom MCU board accepting 24–36V input, stepped down to 3.3V using an onboard buck regulator.
- Integrated USB-C for board power/programming with required power, programming, and protection support circuitry.
- Planned PCB layout around regulator placement, decoupling, grounding, and clean 3.3V delivery for embedded hardware.

BallerIo – Real-Time Multiplayer Web Game – Personal Project *Jul 2026 – Jul 2026*
- Built a real-time multiplayer browser game with Node.js and WebSockets, syncing player state across concurrent clients.
- Implemented merge-attraction mechanics, split/eject actions, and an XP/leveling system with ~30 Upd/S state broadcasts.
- Deployed to production on Render with version control and continuous deployment through GitHub.

Remote-Control Robot – Personal Project *Jan 2026 – Feb 2026*
- Combined RC car and robot concepts into a mobile platform using microcontroller control, dual DC motors/ESCs, sensors, and remote input.
- Wired power distribution and common-ground connections across motors, controllers, sensors, and feedback subsystems.
- Programmed and debugged drive logic, throttle response, obstacle sensing, and button-triggered behaviors to improve reliability under load.

Automated Bridge – Academic Project *Apr 2026 – May 2026*
- Built a motorized bridge prototype that opens/closes automatically using sensor input, traffic LEDs, and Arduino control logic.
- Integrated mechanical supports, bridge movement, and electronics to simulate safe boat/vehicle crossing behavior.

Door Alarm System – Personal Project *Jan 2026 – Feb 2026*
- Built an ESP32-based door alarm with a reed switch, DFPlayer Mini, microSD card, and speaker to play custom audio alerts.
- Implemented UART communication and sensor-state logic to trigger MP3 playback when door/latch movement was detected.

INVOLVEMENTS

UNC Charlotte Robotics Fighting Club *Jul 2025 – Present*
- Participated in robotics meetings and technical discussions related to robot design, mechanical systems, and electronics.